

Abstract Mapping Guide

This document explains what the abstract visualisation is showing.

Overall Reading

The mesh is a dynamic surface driven by audience physiology and behavioural data:

- **Shape** shows slow bodily pressure and temperature movement.
- **Pulse** shows heart rhythm and rhythm instability.
- **Sparks and sudden rings** show skin conductance responses.
- **Colour** shows emotional valence.
- **Motion intensity** shows arousal.
- **Solidity and coherence** shows engagement.

The physiological inputs are standard-deviation signals. In practice, they describe how much a metric is varying. Values near 0 are calm or baseline. Larger positive or negative values create stronger visual changes. Positive and negative values are used as directions.

Affect Signals

Valence

Valence controls the base colour of the mesh.

- **Low valence:** red/orange tones.
- **Neutral valence:** green/cyan tones.
- **High valence:** yellow/gold tones.

This represents the emotional tone of the audience response.

Arousal

Arousal controls how active the mesh feels.

- **Low arousal:** slower, smoother, calmer movement.
- **High arousal:** faster motion, more turbulence, sharper changes, stronger trails.

This represents audience activation or intensity.

Engagement

Engagement controls how solid or dispersed the mesh is.

- **Low engagement:** the mesh becomes looser, dimmer, smaller, and more drifting.
- **High engagement:** the mesh becomes brighter, larger, more coherent, and more solid.

This represents how strongly the audience is collectively held by the performance.

Physiological Standard Deviations

Tonic Electrodermal Activity Standard Deviation

This controls the slow, overall pressure of the mesh.

- **Higher positive values** compress the mesh inward and lift its centre upward.
- **Lower negative values** expand the mesh outward and let it sink or relax downward.

This represents slow bodily tension or release.

Temperature Rate-of-Change Standard Deviation

This controls contour waves across the surface.

- **Larger magnitude** creates taller, more frequent surface contours.
- **Positive movement** appears as warm red/orange wave activity moving from left to right.
- **Negative movement** appears as cool blue wave activity moving from right to left.
- Strong changes can trigger short bursts of travelling colour waves.

This represents thermal change becoming a visible wave or wash through the audience body.

Skin Conductance Response Frequency Standard Deviation

This controls sudden events: rings, sparks, and local shocks in the mesh.

- **Higher magnitude** creates more frequent and stronger events.
- **Positive events** push upward and outward.
- **Negative events** pull inward or downward.
- Strong events brighten nearby particles and can create white spark activity.

This represents moments of audience reaction, surprise, stress, or heightened attention.

Heart Rate Standard Deviation

This controls the rhythmic pulse layer.

- **Positive values** quicken the pulse rate.
- **Negative values** slow the pulse rate.
- **Larger magnitude** makes the breathing/pulse layer more visible and energetic.

This represents the visible tempo of audience physiology.

Inter-Beat Interval Standard Deviation

This controls how regular or broken the pulse rings feel.

- **Positive values** make pulse rings broader and more spacious.
- **Negative values** make pulse rings tighter, more segmented, and more broken.

This represents rhythmic stability versus fragmentation.